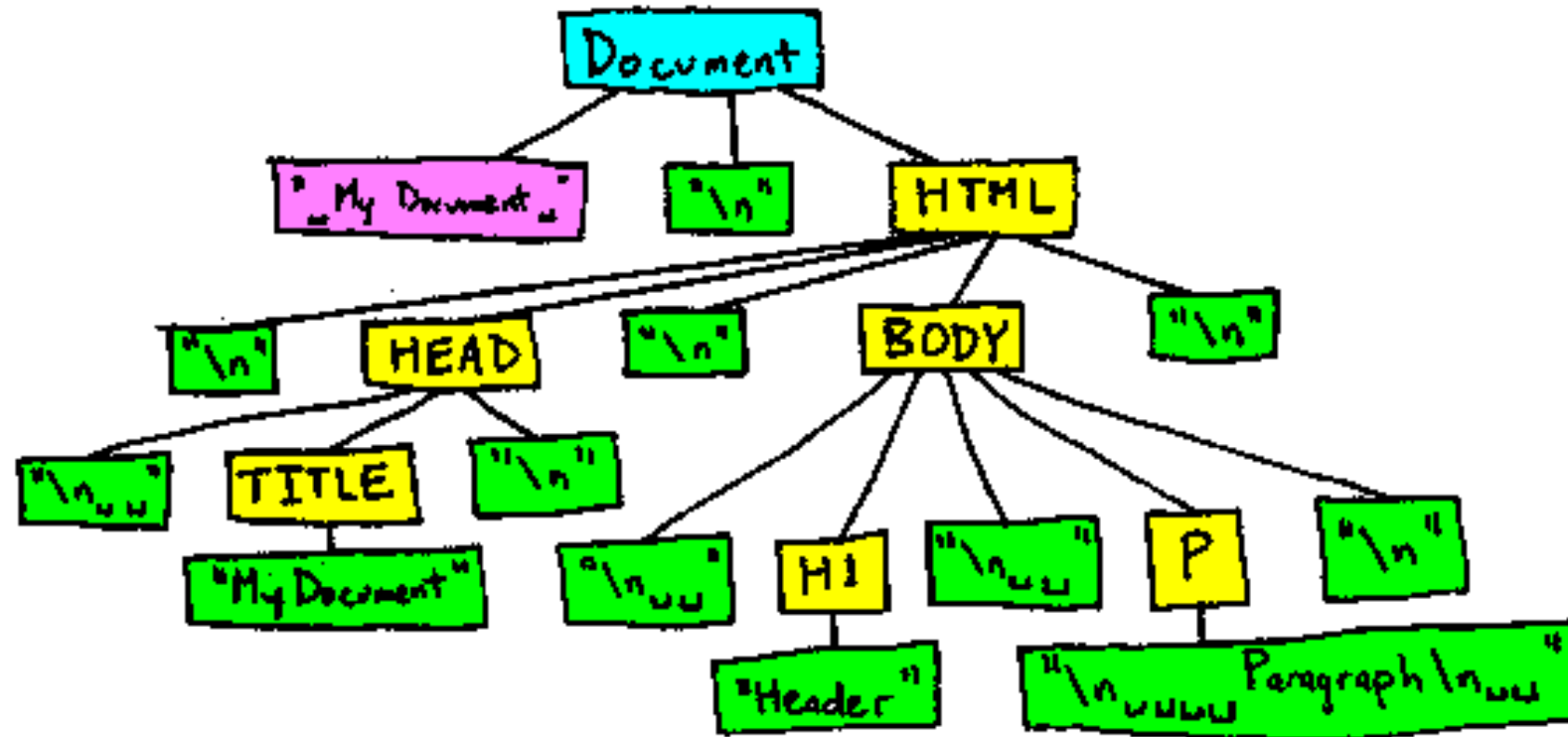


Technologies for E-Commerce

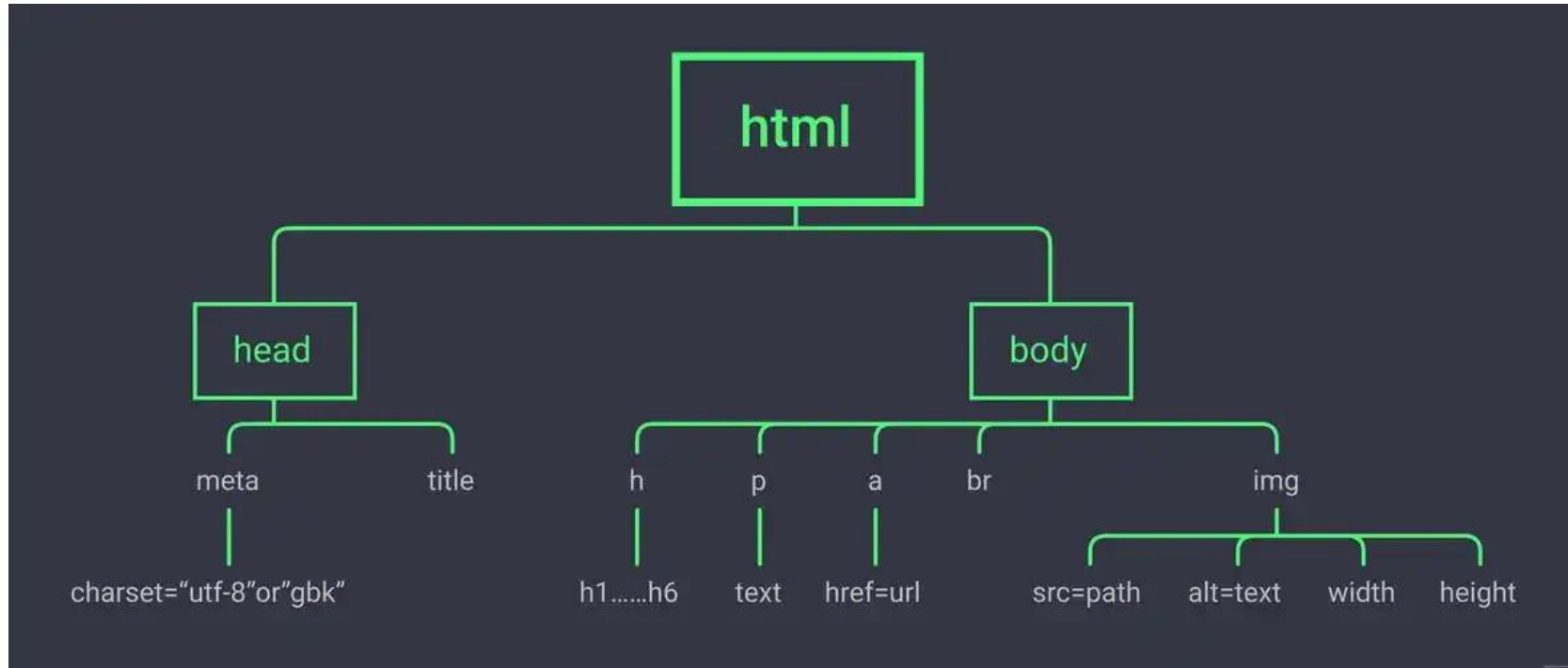
CAN302

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Lecture 2 – Insight into HTML & CSS



HTML – Hyper Text Markup Language



- HTML structures the web page.
- HTML organizes all element as a tree called **Document Object Model (DOM) Tree**

A HTML sample

```
<!DOCTYPE html>
<html>
<head>
<title>Page Title</title>
</head>
<body>

<h1>This is a heading</h1>
<p>This is a paragraph.</p>
<p>This is another paragraph.</p>

</body>
</html>
```

- The <!DOCTYPE html> declaration defines that this document is an **HTML5** document, itself is not a HTML tag
- The <html> element is the root element of an HTML page
- The <head> element contains meta information about the HTML page
- The <title> element specifies a title for the HTML page (which is shown in the browser's title bar or in the page's tab)
- The <body> element defines the document's body, and is a container for all the visible contents, such as headings, paragraphs, images, hyperlinks, tables, lists, etc.
- The <h1> element defines a large heading
- The <p> element defines a paragraph

HTML structure

```
<!DOCTYPE html>
<html>
<head>
<title>Page Title</title>
</head>
<body>

<h1>This is a heading</h1>
<p>This is a paragraph.</p>
<p>This is another paragraph.</p>

</body>
</html>
```

The diagram illustrates the hierarchical structure of the HTML code. It consists of several nested rectangular boxes. The outermost box is labeled `<html>` at the top-left and `</html>` at the bottom-right. Inside this box, there are two main sections: `<head>` and `<body>`. The `<head>` section contains a single box labeled `<title>Page title</title>`. The `<body>` section contains three stacked boxes: the top one is labeled `<h1>This is a heading</h1>`, the middle one is labeled `<p>This is a paragraph.</p>`, and the bottom one is labeled `<p>This is another paragraph.</p>`. This visual representation shows how the tags are nested, with the `<body>` tag enclosing the content tags, which in turn enclose their respective text.

```
<html>
<head>
  <title>Page title</title>
</head>
<body>
  <h1>This is a heading</h1>
  <p>This is a paragraph.</p>
  <p>This is another paragraph.</p>
</body>
</html>
```

- The different tags are nested together.

DOCTYPE

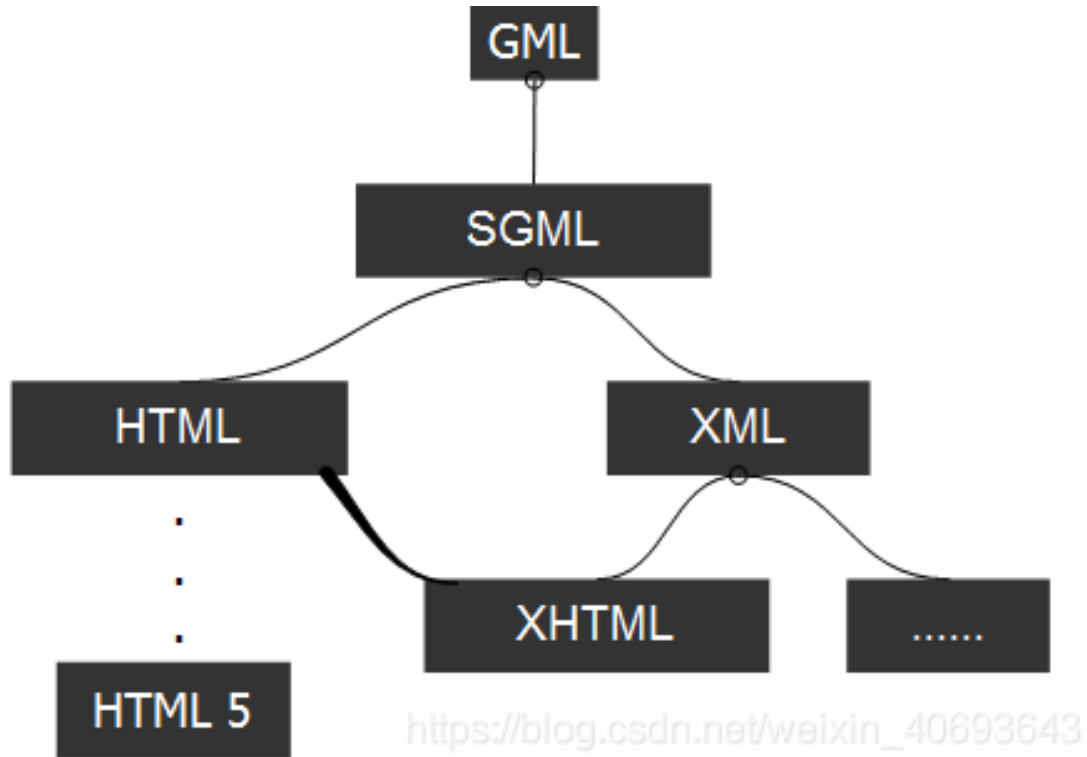
- A brief history of HTML

Year	Version
1989	Tim Berners-Lee invented www
1991	Tim Berners-Lee invented HTML
1993	Dave Raggett drafted HTML+
1995	HTML Working Group defined HTML 2.0
1997	W3C Recommendation: HTML 3.2
1999	W3C Recommendation: HTML 4.01
2000	W3C Recommendation: XHTML 1.0
2008	WHATWG HTML5 First Public Draft
2012	<u>WHATWG HTML5 Living Standard</u>
2014	<u>W3C Recommendation: HTML5</u>
2016	W3C Candidate Recommendation: HTML 5.1
2017	<u>W3C Recommendation: HTML5.1 2nd Edition</u>
2017	<u>W3C Recommendation: HTML5.2</u>

- Different version has different capabilities.
- New features of HTML5 (or **H5**):
 - Video/Audio support: <video> <audio> tag can directly refer to multimedia. Previous web need to install Browser plugin i.e. **Flash**
 - Canvas: <canvas> tag can create a place to draw things. Many web game using this feature.
 - WebSocket: upgrade the communication to dual-direction
 - Access to **local devices**: Camera, location information, sensors...

<https://www.geeksforgeeks.org/top-10-new-features-of-html5/>

HTML revolution



- GML is the 1st generation – General markup language.
- SGML try to make it to be Standard GML.
- HTML or Hyper Text Markup Language is relatively loose and XML or Extensible Markup Language is strict. The difference is DTD or Document Type Definition
- XML is widely used in communication
- XHTML is a conjunction of HTML and XML
- HTML5 is no longer a member of GML

HTML attributes

```
<html xmlns="http://www.w3.org/1999/xhtml" lang="en" xml:lang="en">
```

```
...
```

```
</html>
```



```
<html lang="en">
```

```
...
```

```
</html>
```

- xmlns is XHTML namespace, for XHTML version
- Nowadays, by default, html uses the HTML5 version
- **lang** used to set the language of the whole web page. It can be language only or with the country code, i.e.:
 - zh, en
 - zh-CN, en-US
- You can also set different language from the whole page for a specific zone (tags)

HTTP Head vs. HTML Head

```
HTTP/1.1 200 OK
```

```
Date: Sun, 08 Feb xxxx 01:11:12 GMT
```

```
Server: Apache/1.3.29 (Win32)
```

```
Last-Modified: Sat, 07 Feb xxxx
```

```
ETag: "0-23-4024c3a5"
```

```
Accept-Ranges: bytes
```

```
Content-Length: 35
```

```
Connection: close
```

```
Content-Type: text/html
```

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<title>Page Title</title>
```

```
</head>
```

```
<body>
```

```
<h1>This is a heading</h1>
```

```
<p>This is a paragraph.</p>
```

```
<p>This is another paragraph.</p>
```

```
</body>
```

```
</html>
```

- They are totally different!
- HTTP head is the information of HTTP request/response, or the communication between C/S.
- The whole HTML file is a part of the HTTP response!
- HTML head is a part of the HTML file.

HTML – within the head

- `<title>`
 - All elements in **head** is **NOT** visible on webpage...
 - **title** is shown in browser tab, bookmark and search result
- `<meta>`
 - **meta** or Metadata is data that describes data, and HTML has an "official" way of adding metadata to a document. There are many different metadata, such as `<meta charset="utf-8" />` for characters encoding.
- `<style>`
 - **style** is used to refer to the css files
- `<link>`
 - **link** is used to have an icon
- `<script>`
- `<base>`

HTML – within the head

- `<title>`
- `<meta>`
- `<style>`
- `<link>`
- `<script>`
 - `script` is used to refer to the JavaScript need to be loaded, as:
`<script src="my-js-file.js" defer></script>`
 - It can refer to the JS source on any accessible web source.
 - The attribute "defer" should be used that load the webpage first, then load the JS at last. Otherwise the JS may block the webpage loading.
 - But not every browser support the "defer", so it is better to put all `<script>` tag at the bottom of `<body>` to guarantee the JS was loaded at last.
- `<base>`
 - `base` specifies the base URL to use for all relative URLs in a document.

HTML – tags in body

div	division	分割 (块元素)
span	span	区域 (行内元素)
p	paragraph	段落
ol	ordered list	有序列表
ul	unordered list	无序列表
li	list item	列表项
dl	definition list	定义列表
dt	definition term	定义术语
dd	definition description	定义描述
h1~h6	header1 ~header6	标题1~标题6
hr	horizontal rule	水平线
a	anchor	锚点, 超链接
strong	strong	强调 (粗体)
em	emphasized	强调 (斜体)
sup	superscripted	上标
sub	subscripted	下标
br	break	换行
fieldset	fieldset	域集
legend	legend	图例
caption	caption	(表格、图像等) 标题
thead	table head	表头
tbody	table body	表身
tfoot	table foot	表脚
th	table header	表头单元格
td	td	表身单元格

- HTML is **very easy** to learn
- HTML is **difficult** to be a master
- You may get the webpage with similar visual effect by using different tags
- But you are strongly recommended to use the “**correct**” tags according to the purpose of such tags.

https://zhuanlan.zhihu.com/p/500559740?utm_id=0

HTML – paragraph

<code><h1>~<h6></code>	header
<code><p></code>	paragraph
<code>
</code>	break
<code><hr></code>	horizontal rule
<code><div></code>	division
<code></code>	span

- Header **again**? This head is not the one for whole page but the one shown on the webpage. Head for human beings to look.
- Why we need some many different tags?
- Use `<div>` with different style, we can fulfill both header and paragraph.
- While many “users” is not human but computer! They need to read such tags.
- NLP may can read the webpage more like human beings.

HTML – paragraph

<code></code>	strong (加强)
<code></code>	emphasized (强调)
<code><cite></code>	cite (引用)
<code><sup></code>	superscripted (上标)
<code><sub></code>	subscripted (下标)

HTML

`</>` Play 

```
<p>
```

```
    Before proceeding, <strong>make sure  
you put on your safety  
goggles</strong>.
```

```
</p>
```

- Similar function to the **strong** by Microsoft PowerPoint!

<https://developer.mozilla.org/en-US/docs/Web/HTML/Element/strong>

HTML – list

- List tags

ol	ordered list
ul	unordered list
dl	definition list

HTML

```
<ul>
  <li>first item</li>
  <li>second item</li>
  <li>third item</li>
</ul>
```

- Similar function to the list function of Microsoft office!

<https://developer.mozilla.org/en-US/docs/Web/HTML/Element/ul>

HTML – table

- Table tags

table	table (表格)
tr	table row (表格行)
td	table data cell (表格单元格)

	<td>	<td>	<td>
<tr>	Content	Content	Content
<tr>	Content	Content	Content
<tr>	Content	Content	Content

- Similar to the table of Microsoft office!

<https://ictacademy.com.ng/html-tables/>

HTML – table

- Table tags

thead	table head
tbody	table body
tfoot	table foot
th	table header

Title	Price	Available
CSS Demystified	\$29	16
Mastering JavaScript	\$35	10
HTML5: An Introduction	\$15	6
Total		32

```
<table>
  <thead>
    <tr>
      <th> Title </th>
      <th> Price </th>
      <th> Available </th>
    </tr>
  </thead>
  <tbody>
    <tr>
      <td> CSS Demystified </td>
      <td> $29 </td>
      <td> 16 </td>
    </tr>
    <tr>
      <td> Mastering JavaScript </td>
      <td> $35 </td>
      <td> 10 </td>
    </tr>
    <tr>
      <td> HTML5: An Introduction </td>
      <td> $15 </td>
      <td> 6 </td>
    </tr>
  </tbody>
  <tfoot>
    <tr>
      <td> Total </td>
      <td></td>
      <td> 32 </td>
    </tr>
  </tfoot>
</table>
```

- Similar to the table of Microsoft office!

https://zhuanlan.zhihu.com/p/500559740?utm_id=0

HTML – table

- **td attribute**

colspan

rowspan

```
<table width="200" height="200" border="2" bgcolor="yellow">
```

```
<tr>
```

```
<td colspan="2">1</td>
```

```
<td>3</td>
```

```
</tr>
```

```
<tr>
```

```
<td rowspan="2">4</td>
```

```
<td>5</td>
```

```
<td>6</td>
```

```
</tr>
```

```
<tr bgcolor="red">
```

```
<td>8</td>
```

```
<td>9</td>
```

```
</tr>
```

```
</table>
```

“colspan” is used to
combine columns .

“rowspan” is used to
combine rows .

Pay attention to cell “2” &
“7”, they have been deleted
as “1” & “4” have taken
their place .

Here is the result .

1		3
4	5	6
	8	9

- Similar to the table of your Microsoft office!

HTML – image

HTML Demo: RESET

HTML	CSS	OUTPUT
<pre>1 5</pre>		

- src to specify the image's path
- alt used to show the text if the img cannot be loaded by the browser

HTML – hyperlink

HTML Demo: <a> RESET

HTML	CSS	OUTPUT
<pre>1 <p>You can reach Michael at:</p> 2 3 4 Websi te 5 Email 6 Phone </pre>		You can reach Michael at: • Website • Email • Phone

- href to tell the browser for the URL once it is clicked!

<https://developer.mozilla.org/en-US/docs/Web/HTML/Element/a>

HTML – form

Input

Select

Textarea

Multiselect

Checkboxes

☒ Option 1
☐ Option 2
☐ Option 3

Radio Buttons

☒ Option 1
☐ Option 2
☐ Option 3

- `<form>` represents a document section containing interactive controls for **submitting** information.
- There are four types of tags:
 - `<label>`
 - `<input>`
 - `<textarea>`
 - `<select>` + `<option>`

https://zhuanlan.zhihu.com/p/500559740?utm_id=0

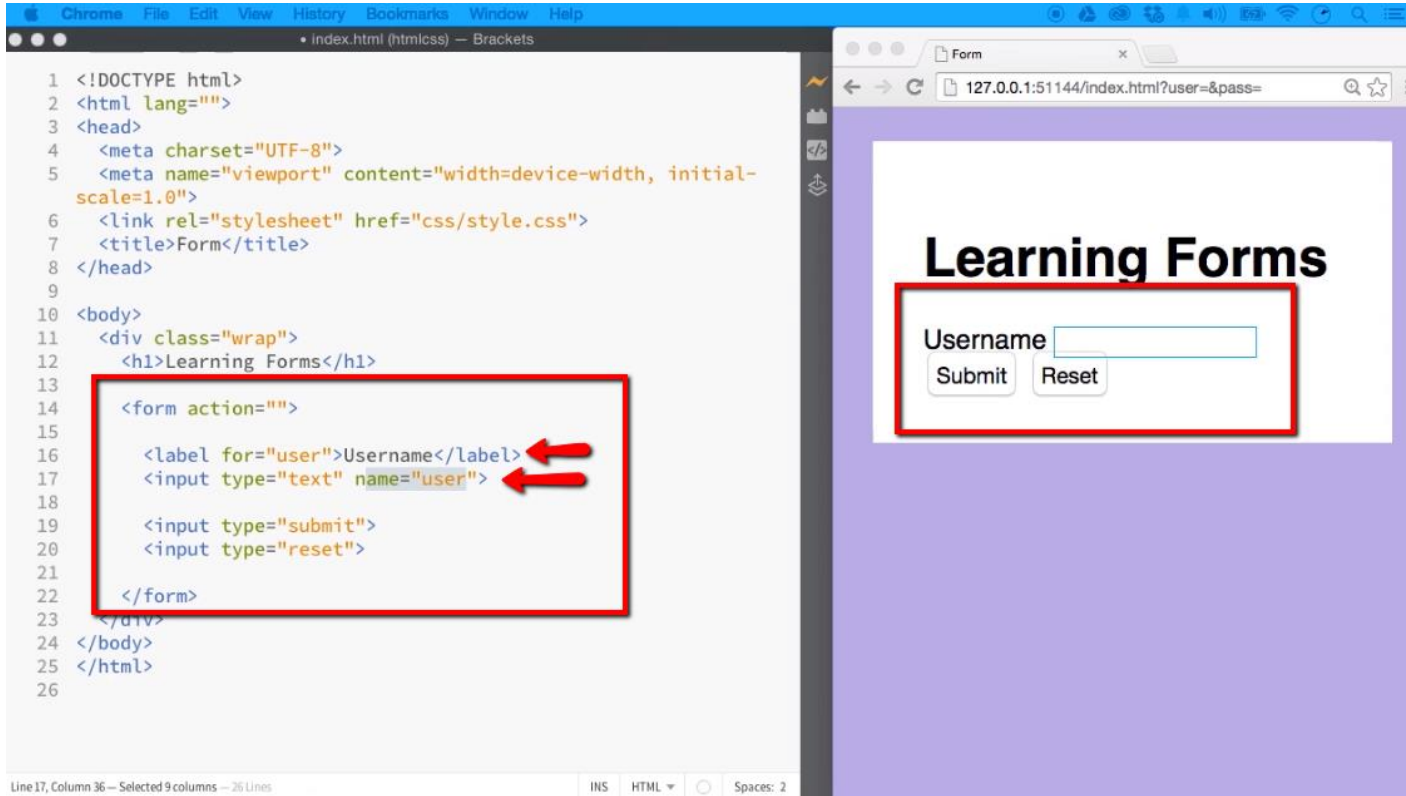
HTML – form

```
<form action="/action_page.php" method="get">
  <label for="fname">First name:</label>
  <input type="text" id="fname" name="fname"><br><br>
  <label for="lname">Last name:</label>
  <input type="text" id="lname" name="lname"><br><br>
  <input type="submit" value="Submit">
</form>
```

- “action” defined an target URL to submit the data in the form
- “method” defined the http request method, it could be GET or POST

https://www.w3schools.com/tags/att_form_action.asp

HTML – form - label



- `<label>` associates with a form control, such as `<input>` or `<textarea>` offers some major advantages:
- The label text is not only visually associated with its corresponding text input; it is programmatically associated with it too
- When a user clicks or touches/taps a label, the browser passes the focus to its associated input

<https://developer.mozilla.org/en-US/docs/Web/HTML/Element/label>

HTML – form - input

Input Type	Description
email	For entering an email address
tel	For entering a telephone number
url	For entering a URL
search	a text input field with line breaks automatically removed
number	For entering a floating point number
range	For entering an approximate value; represented by a slider
date	Enter the day, month, and year
datetime	For entering the day, month, year, hour, minute, second and microsecond based on the current timezone
datetime-local	For entering a date and time with no timezone
month	For entering the month and year with no timezone
week	For entering a week number with no timezone
time	For entering the time with no timezone
color	For specifying a color

```
<input type="text">
<input type="email">
<input type="password">
<input type="checkbox">
<input type="date">
<input type="color">
<input type="button">
<input type="range">
<input type="radio">
<input type="url">
<input type="reset">
<input type="datetime-local">
<input type="month">
<input type="week">
<input type="tel">
<input type="file">
<input type="hidden">
<input type="image">
<input type="search">
<input type="number">
```

Hello World

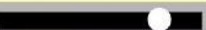
e-mail address

☐ ☒

02-10-23 



Button



No ☐ Yes

https://abc.com

Reset

02-10-23 10:11 

Feb 2023 

Week 6, 2023 

9999000088

Choose File



Submit Image

Search

9

HTML – form – textarea

Example (Text Area)	It looks like this in the browser
<pre>your comments:
 <textarea name="TA1" cols="40" rows="8" maxlength="100" placeholder="type only 100 characters" required> </textarea></pre>	 A screenshot of a web browser window. The title bar says 'Your Comments'. Inside the browser, there is a text area with a light gray border. The text area contains the placeholder text 'type only 100 characters' in a light blue font. The text area is rectangular and occupies most of the browser window.

- <textarea> is for mutil-lines input

HTML – form – select + option

HTML Demo: <option>

RESET

HTML	CSS	OUTPUT
<pre>1 <label for="pet-select">Choose a pet:</label> 2 3 <select id="pet-select"> 4 <option value="">--Please choose an option--</option> 5 <option value="dog">Dog</option> 6 <option value="cat">Cat</option> 7 <option value="hamster">Hamster</option> 8 <option value="parrot">Parrot</option> 9 <option</pre>		Choose a pet: ✓ --Please choose an option-- Dog Cat Hamster Parrot Spider Goldfish

- Let user select pre-defined options

https://zhuanlan.zhihu.com/p/500559740?utm_id=0

<https://developer.mozilla.org/en-US/docs/Web/HTML/Element/option>

HTML – form – select + option

HTML Demo: <optgroup> RESET

	HTML	CSS	OUTPUT
1	<code><label for="dino-select">Choose</code>		
	<code>a dinosaur:</label></code>		
2	<code><select id="dino-select"></code>		
3	<code> <optgroup label="Theropods"></code>		
4	<code></code>		
	<code> <option>Tyrannosaurus</option></code>		
5	<code></code>		
	<code> <option>Velociraptor</option></code>		
6	<code> <option>Deinonychus</option></code>		
7	<code> </optgroup></code>		
8	<code> <optgroup label="Sauropods"></code>		
9	<code> <option>Diplodocus</option></code>		
10	<code> <option>Saltasaurus</option></code>		
11	<code> <option>Apatosaurus</option></code>		
12	<code></optgroup></code>		

Choose dinosaur:
Theropods
✓ Tyrannosaurus
Velociraptor
Deinonychus
Sauropods
Diplodocus
Saltasaurus
Apatosaurus

- Options can be organized in groups by <optgroup>

https://zhuanlan.zhihu.com/p/500559740?utm_id=0

<https://developer.mozilla.org/en-US/docs/Web/HTML/Element/optgroup>

HTML – iframe

```
test2.html UNREGISTERED
test2.html x
1 <!DOCTYPE html>
2 <html lang="zh-CN">
3 <head>
4     <meta charset="UTF-8">
5     <meta name="viewport" content="width=device-width, initial-scale=1.0">
6     <title>Document</title>
7 </head>
8 <body>
9
10     <iframe src="https://www.xjtlu.edu.cn" frameborder="3" width="1024" height="400"></
    iframe>
11     <br>
12     <iframe src="https://www.bing.com" frameborder="3" width="1024" height="400"></iframe>
13
14 </body>
15 </html>
```

- Include other pages within the current page

https://zhuanlan.zhihu.com/p/500559740?utm_id=0

HTML – iframe sample

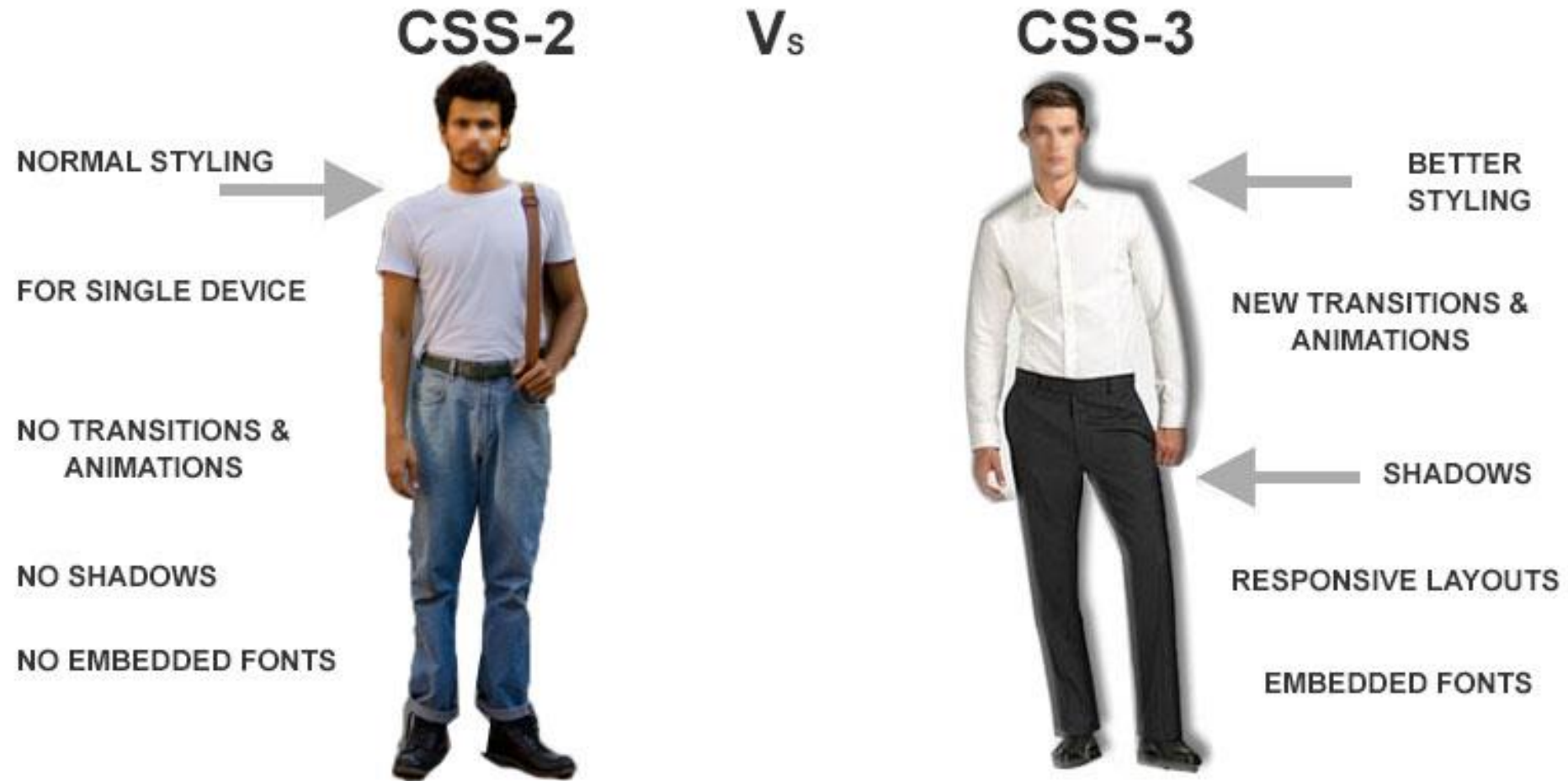


CSS - Cascading Style Sheets



- HTML is a markup language used to format/structure a web page,
- CSS is a **DESIGN** language that you use to make your web page look nice and presentable.

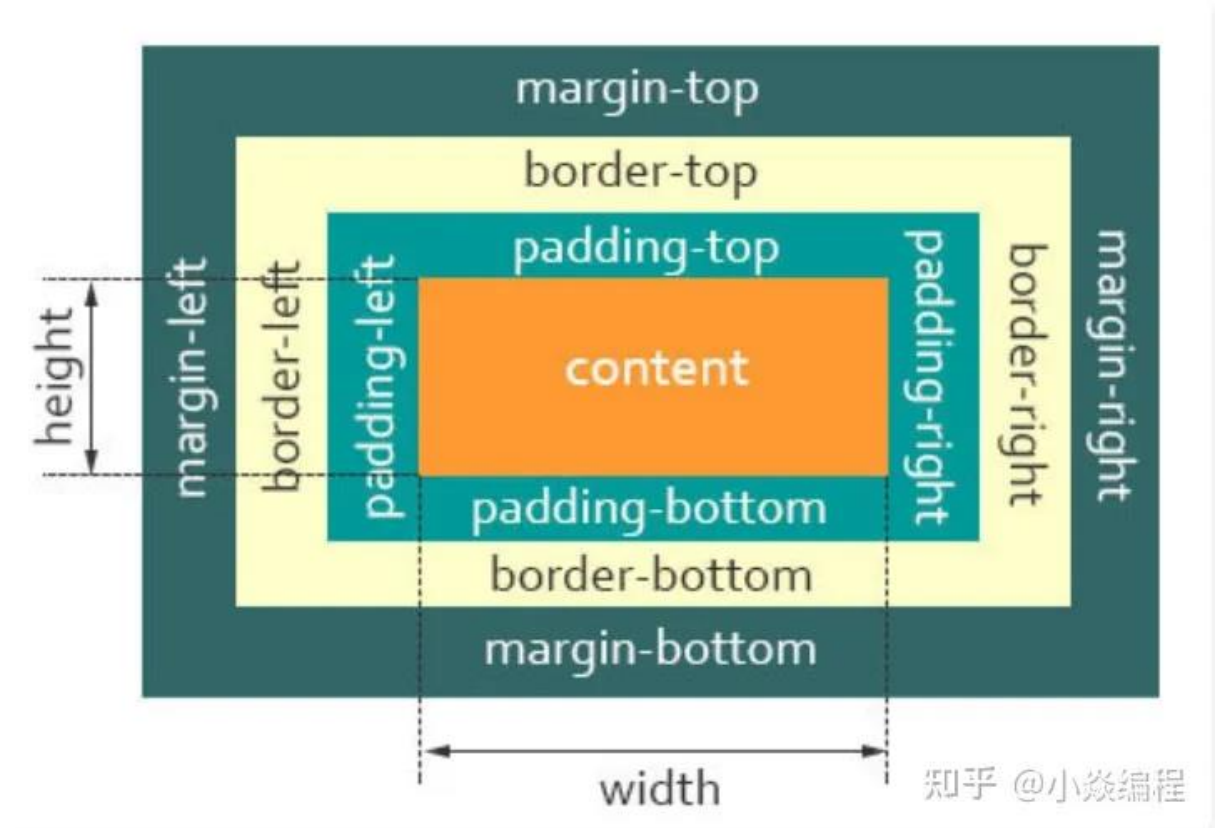
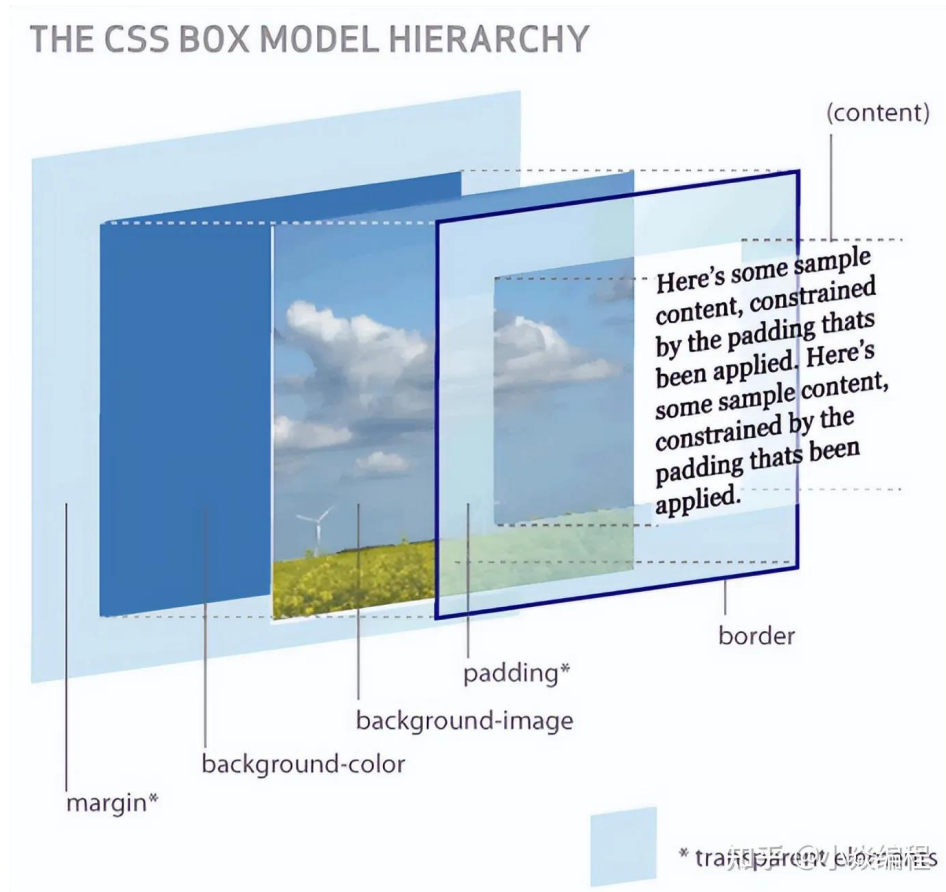
CSS3



- CSS3 is the mainstream version

<https://tutorial.techaltum.com/css3.html>

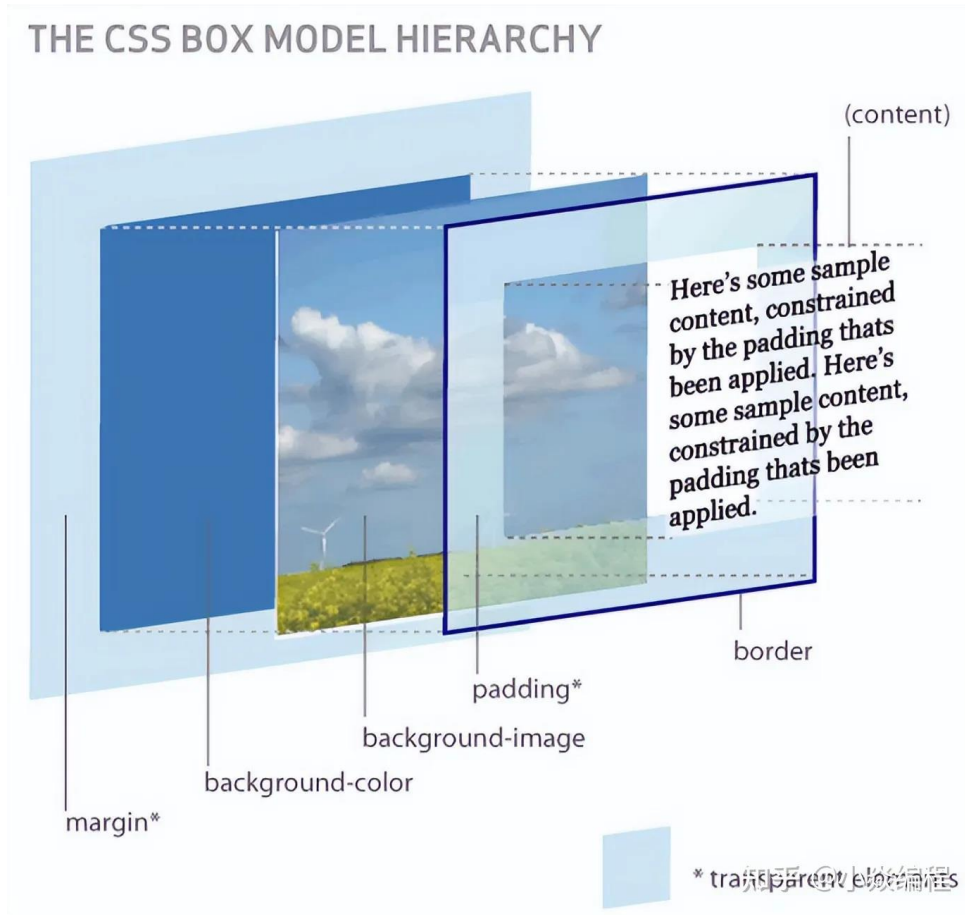
CSS – Box model



<https://zhuanlan.zhihu.com/p/546124925>

https://developer.mozilla.org/en-US/docs/Learn/CSS/Building_blocks/The_box_model

CSS – Box model

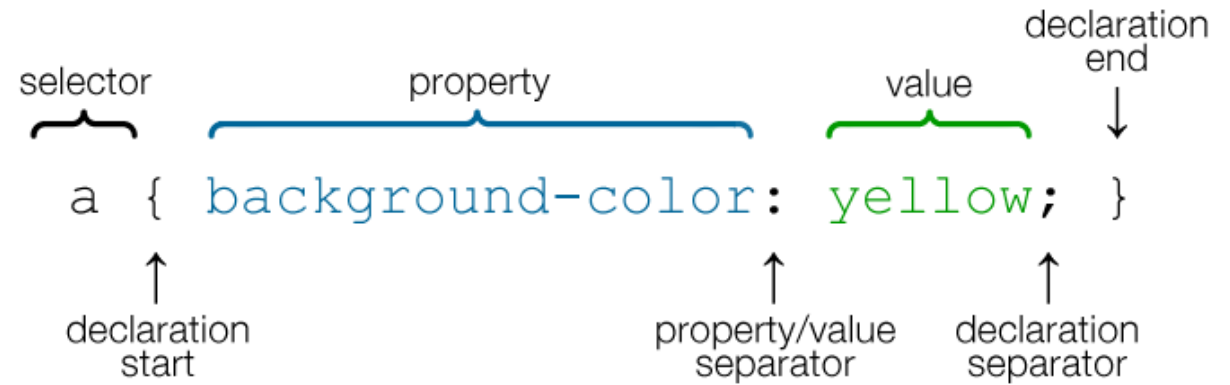


- Background color and image can be set for each element
- Background image should match the size of content. If mismatch, there are several ways to fit them:
 - If box > picture : repeat-x and repeat-y
 - If box < picture: only show part of the picture
- Repeat or not and position can be adjusted

<https://zhuanlan.zhihu.com/p/546124925>

https://developer.mozilla.org/en-US/docs/Learn/CSS/Building_blocks/The_box_model

CSS syntax



- Both style are OK
- Multi-line is easy for human beings to read especially for multi properties.

CSS – **four** different ways to set style for a page

Inline CSS

```
<p style="color: blue;">This is a paragraph.</p>
```

Internal CSS

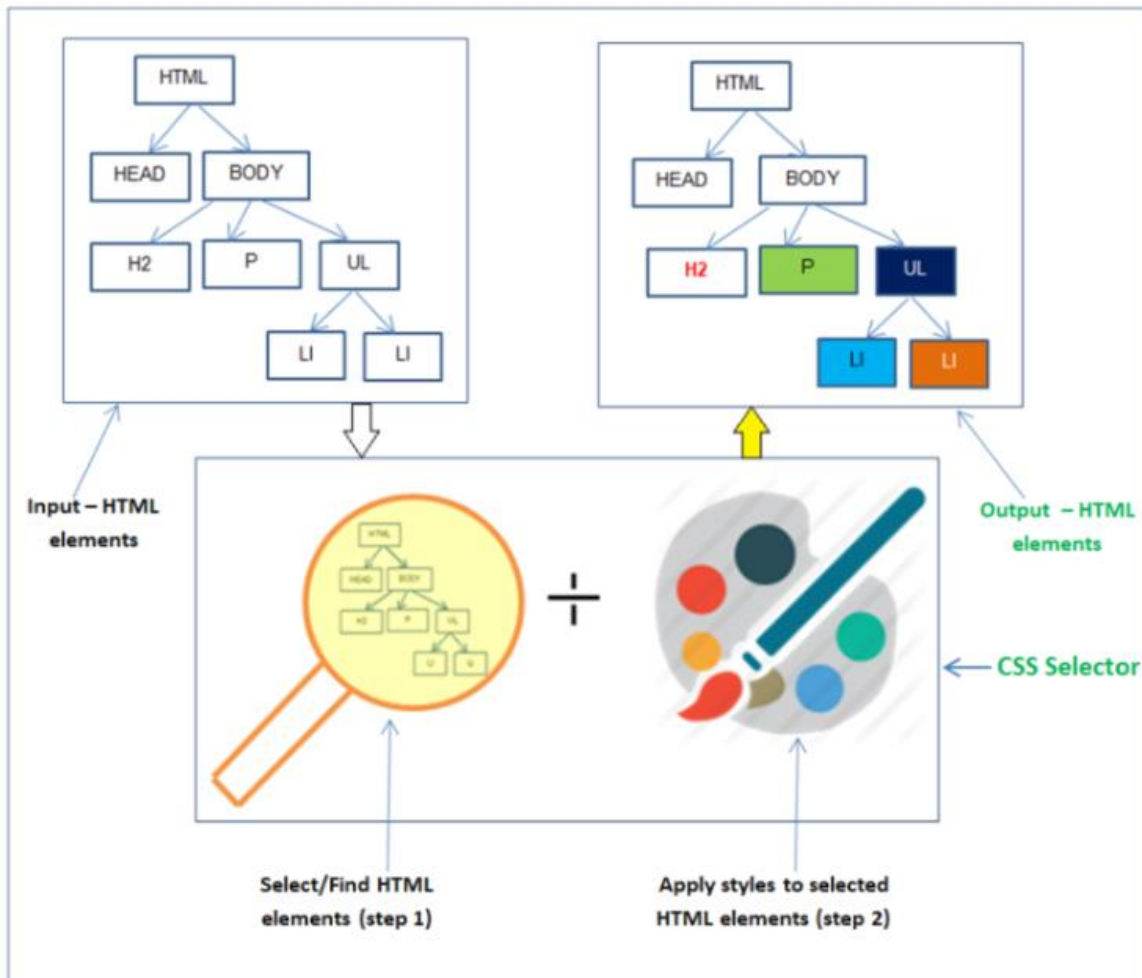
```
<head>
  <style type = text/css>
    body {background-color: blue;}
    p { color: yellow;}
  </style>
</head>
```

External CSS

```
<head>
  <link rel="stylesheet" type="text/css" href="style.css">
</head>
```

- Style is an attribute for HTML element
- Inline means write the style directly with a specific HTML element.
- Internal means the style code is within the HTML code (in the <head> part), these style only effect for this webpage
- External means the style code is in a sperate .css file, it can be referred by different webpages
- **Where** is the fourth one?

CSS selector – link the style to HTML element



- HTML organizes all element as a tree called Document Object Model (DOM) Tree
- Style should be applied to each element of the DOM tree
- Selector is the link between DOM element and style
- **Multi-style may applied to one DOM element**

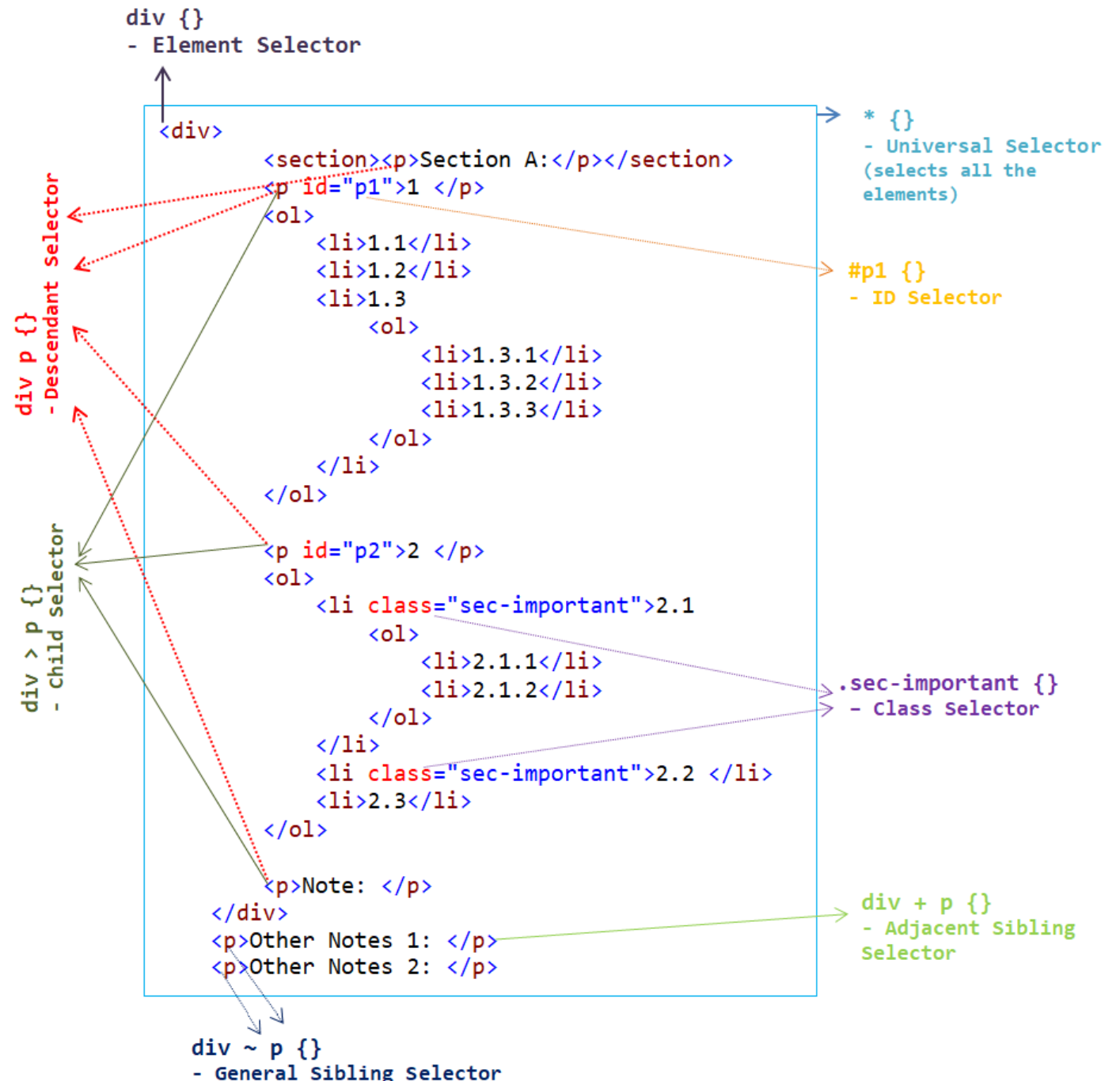
Override: Inline style > Internal CSS > External CSS > Browser default styles.

CSS selector

Selector	Example	Example description	CSS
<u>.class</u>	.intro	Selects all elements with class="intro"	1
<u>#id</u>	#firstname	Selects the element with id="firstname"	1
<u>*</u>	*	Selects all elements	2
<u>element</u>	p	Selects all <p> elements	1
<u>element,element</u>	div,p	Selects all <div> elements and all <p> elements	1
<u>element element</u>	div p	Selects all <p> elements inside <div> elements	1
<u>element>element</u>	div>p	Selects all <p> elements where the parent is a <div> element	2
<u>element+element</u>	div+p	Selects all <p> elements that are placed immediately after <div> elements	2

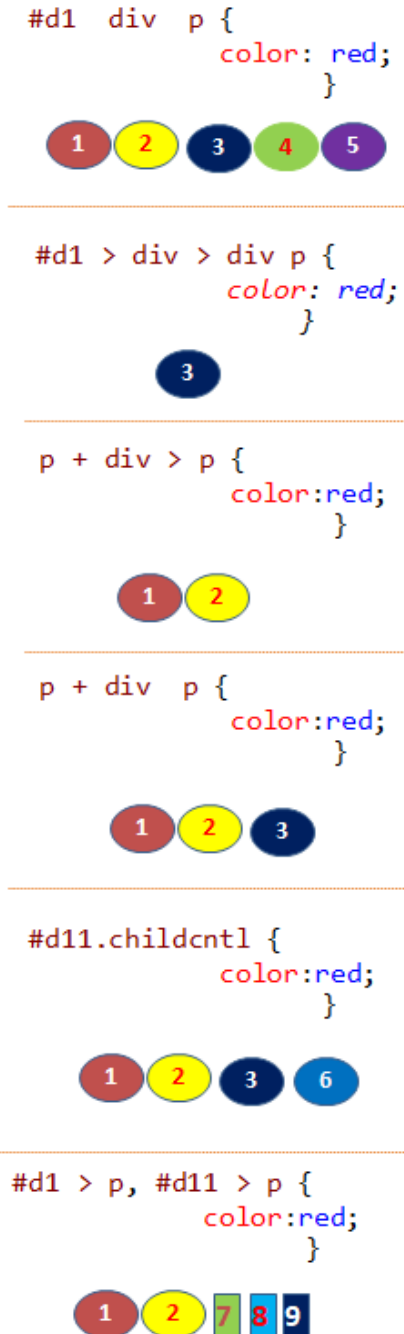
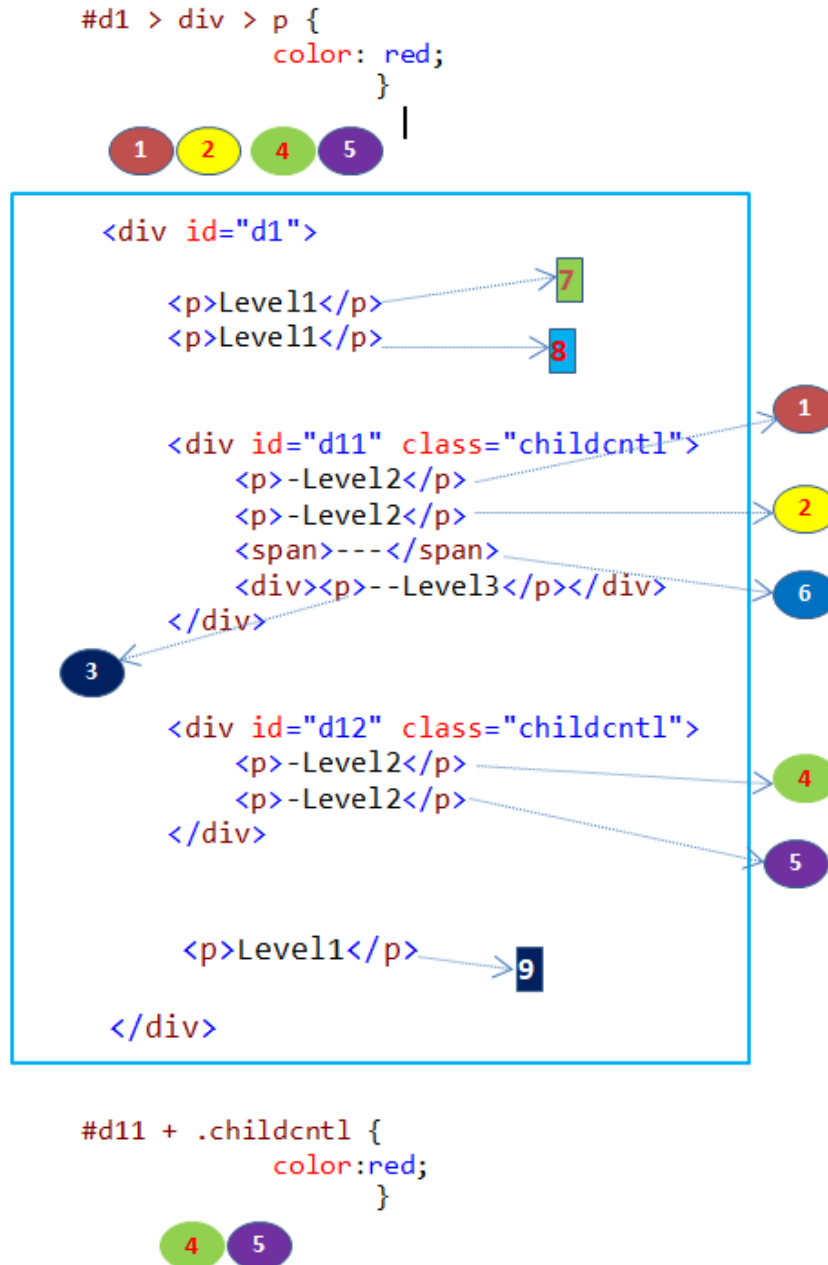
CSS selector

- Universal
- Element
- Class
- ID
- Child
- Descendent
- Adjacent sibling
- General sibling

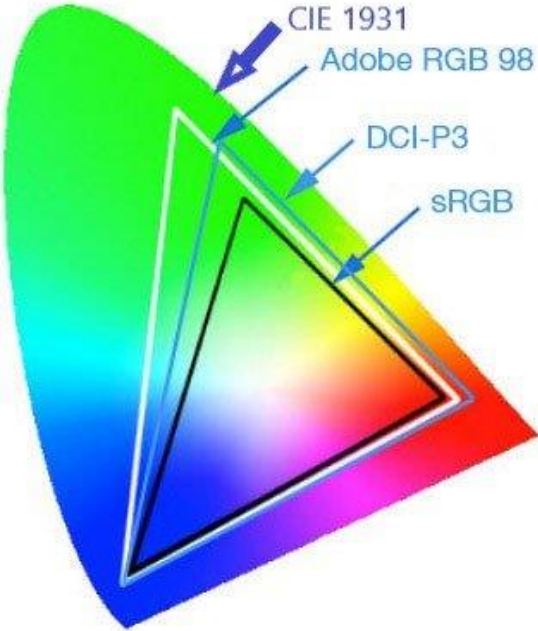
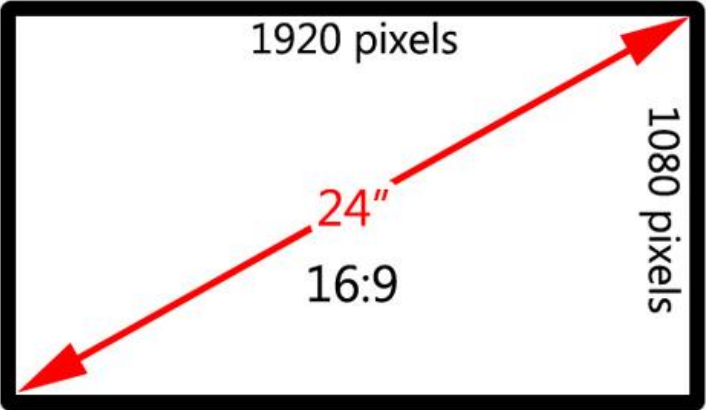
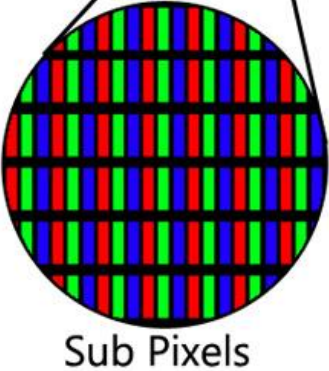
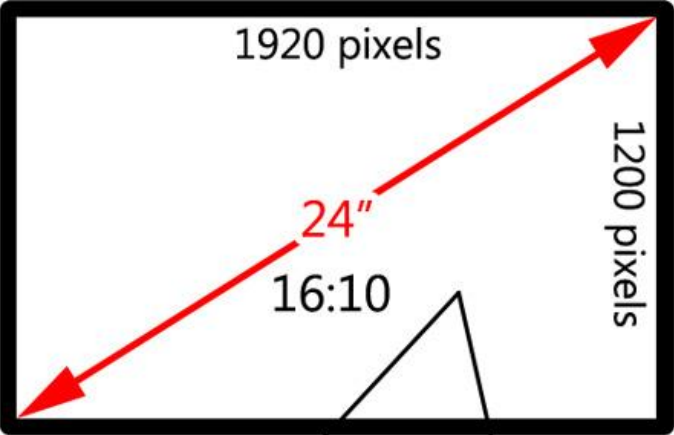


CSS selector

- It is a complex relation between styles to DOM element
- The final result? Please try the code!

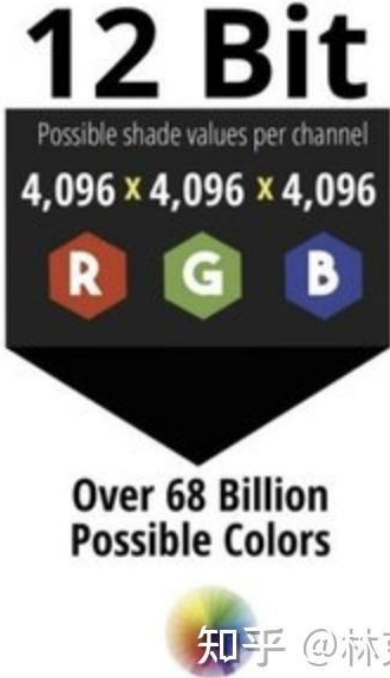


Display mechanism

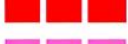
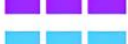


Color Chart	R	G	B	Color Name
	0	0	0	Black
	255	255	255	White
	224	224	224	Light Gray
	128	128	128	Gray
	64	64	64	Dark Gray
	255	0	0	Red
	255	96	208	Pink
	160	32	255	Purple
	80	208	255	Light Blue
	0	32	255	Blue
	96	255	128	Yellow-Green
	0	192	0	Green
	255	224	32	Yellow
	255	160	16	Orange
	160	128	96	Brown
	255	208	160	Pale Pink

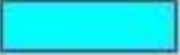
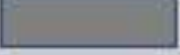
Display mechanism



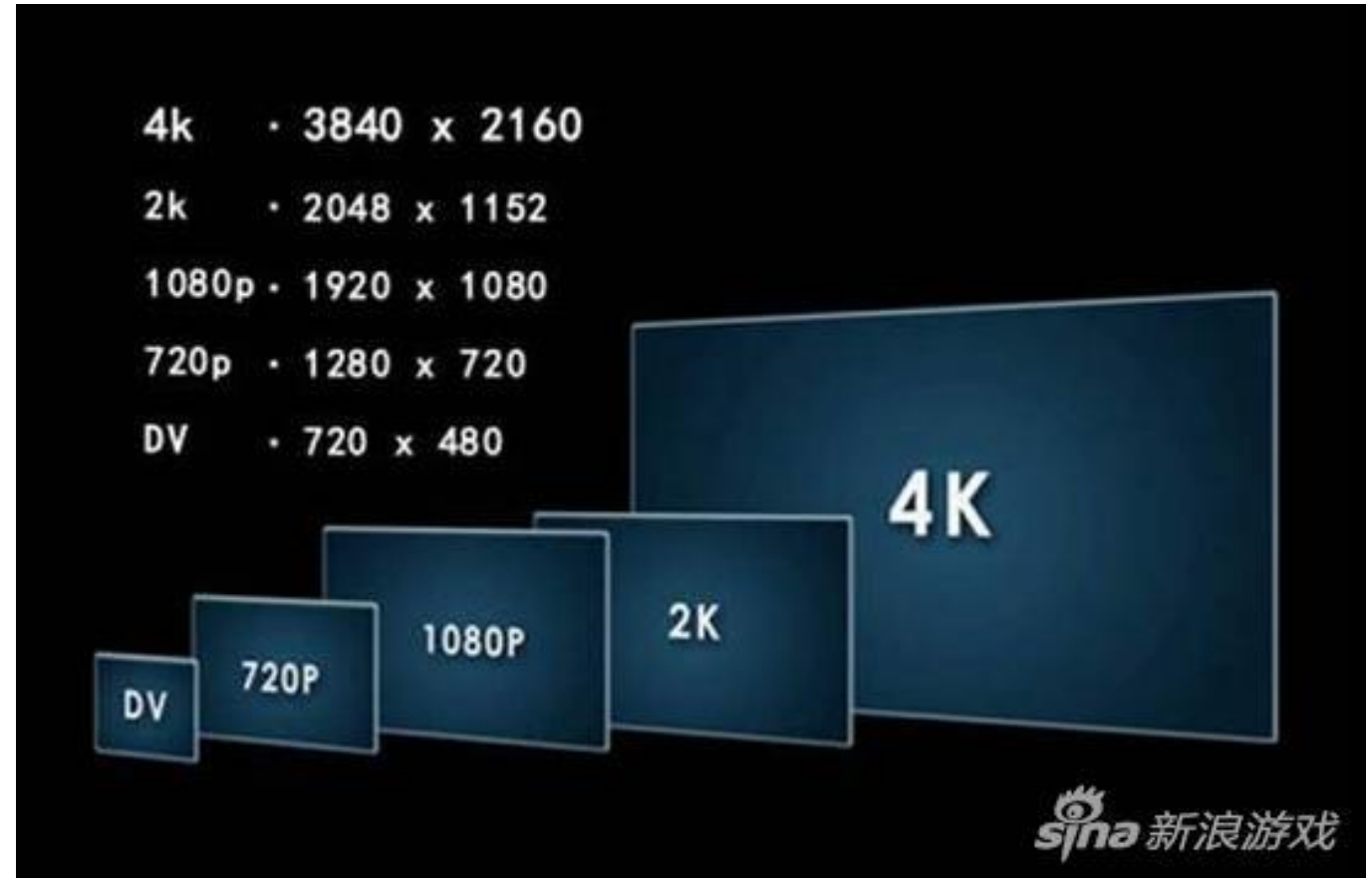
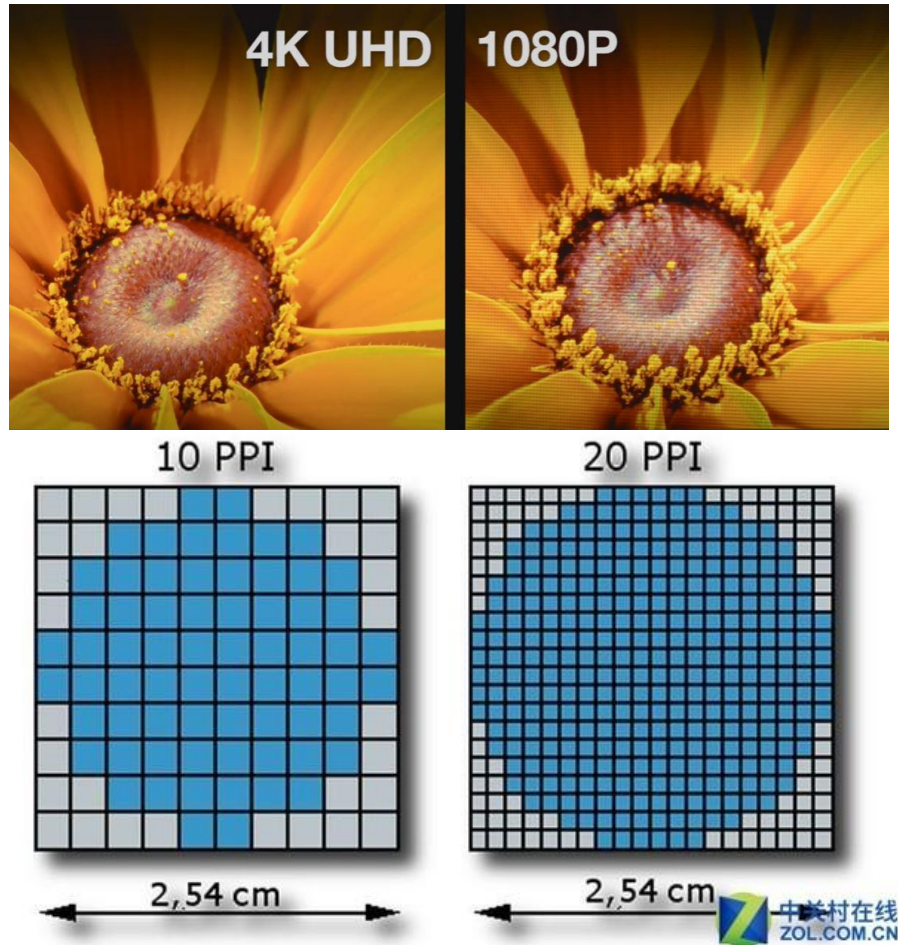
知乎 @林克

Color Chart	R	G	B	Color Name
	0	0	0	Black
	255	255	255	White
	224	224	224	Light Gray
	128	128	128	Gray
	64	64	64	Dark Gray
	255	0	0	Red
	255	96	208	Pink
	160	32	255	Purple
	80	208	255	Light Blue
	0	32	255	Blue
	96	255	128	Yellow-Green
	0	192	0	Green
	255	224	32	Yellow
	255	160	16	Orange
	160	128	96	Brown
	255	208	160	Pale Pink

CSS color

Color	CSS Color Name	Hex Code #RPGGBB	Decimal Code (R,G,B)
	Red	#FF0000	rgb (255,0,0)
	Orange	#FFA500	rgb (255,165,0)
	Yellow	#FFFF00	rgb (255,225,0)
	Green	#008000	rgb (0,128,0)
	Cyan	#00FFFF	rgb (0,255,225)
	Blue	#0000FF	rgb (0,0,225)
	Purple	#800080	rgb (128,0,128)
	Pink	#FFC0CB	rgb (255,192,203)
	Gray	#808080	rgb (128,128,128)
	Brown	#A52A2A	rgb (165,42,42)

Displays resolutions



- Webpage design need to follow the mainstream display resolution

CSS units

Absolute

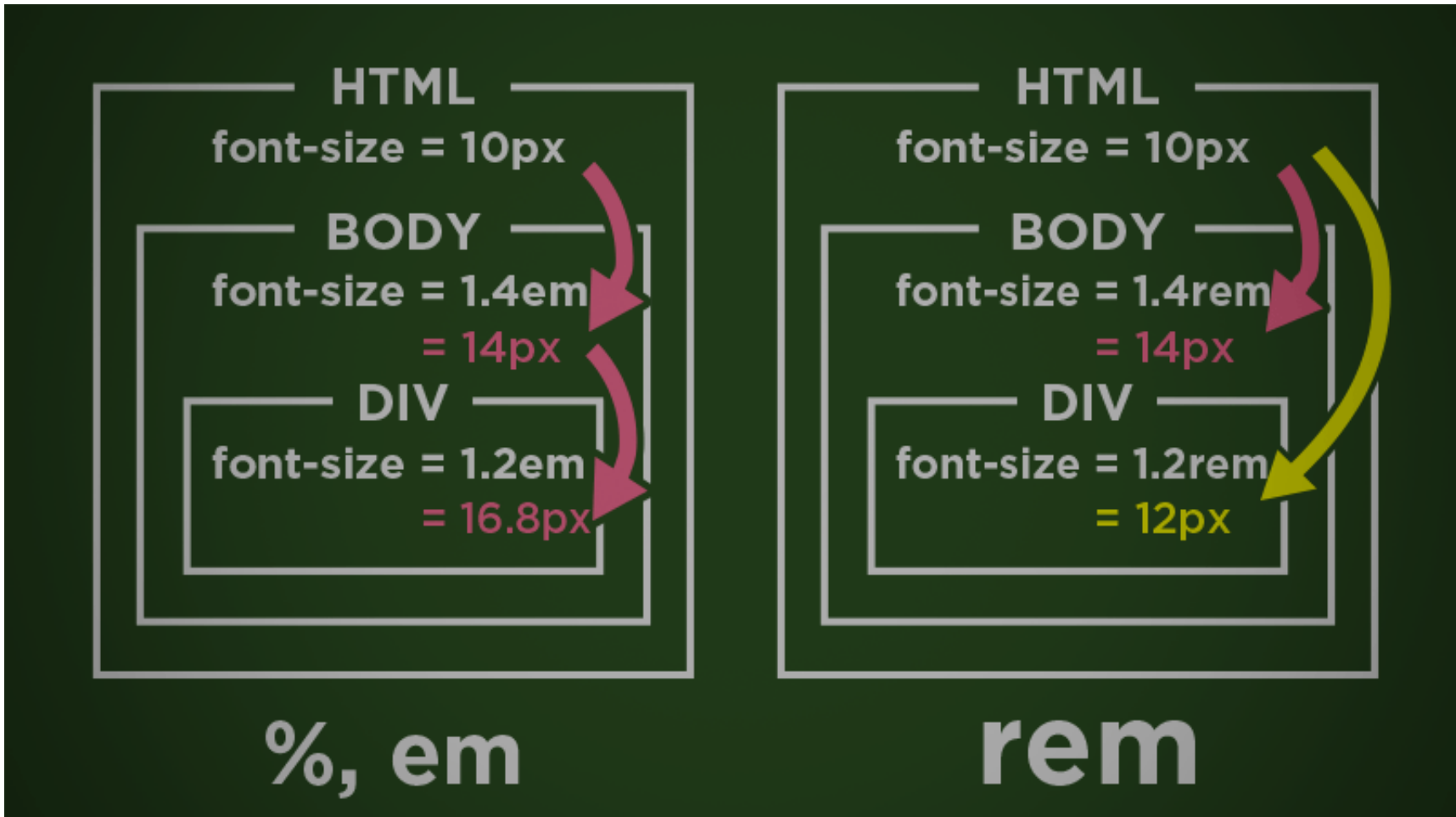
Pixels (px)
Centimeters (cm)
Millimeters (mm)
Inches (in)
Points (pt)
Picas (pc)

Relative

Percentages (%)
Font-sizes (em&rem)
Character-sizes (ex&ch)
Viewport Dimensions (vw &vh)
Viewport Max (vmax)
Viewport Min (vmin)

- **px (such as font-size: 12px) is the unit for pixels.**
- **pt (such as font-size: 12pt) is the unit for points, for measurements typically in printed media.**
- **% (such as width: 80%) is the unit for percentages.**
- **em (such as font-size: 2em) is the unit for the calculated size of a font. So “2em”, for example, is two times the current font size.**

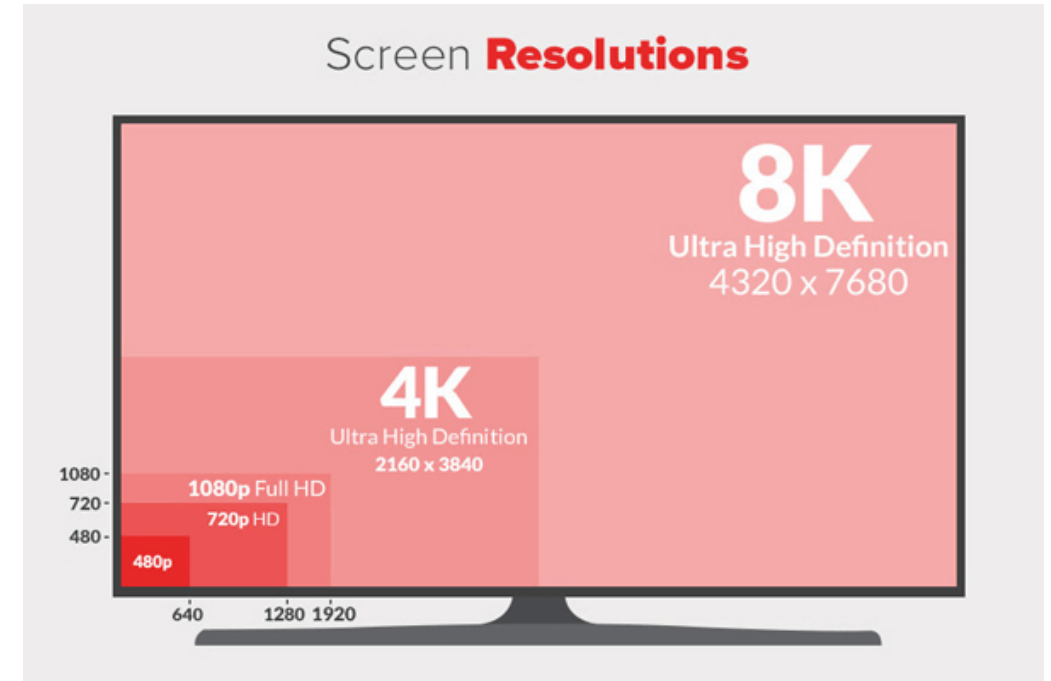
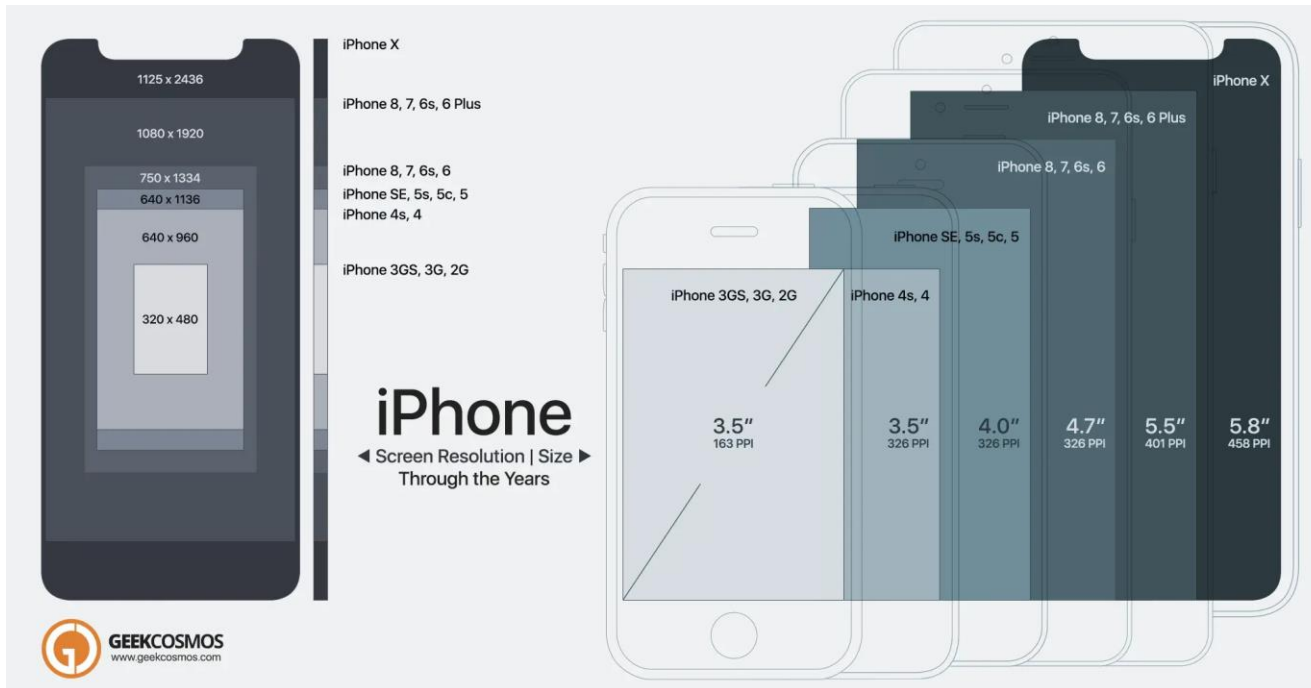
CSS units



- A sample for em and rem

<https://www.pc-koubou.jp/magazine/23912>

Different displays



- Mobile phone is vertical display
- Desktop is horizontal display

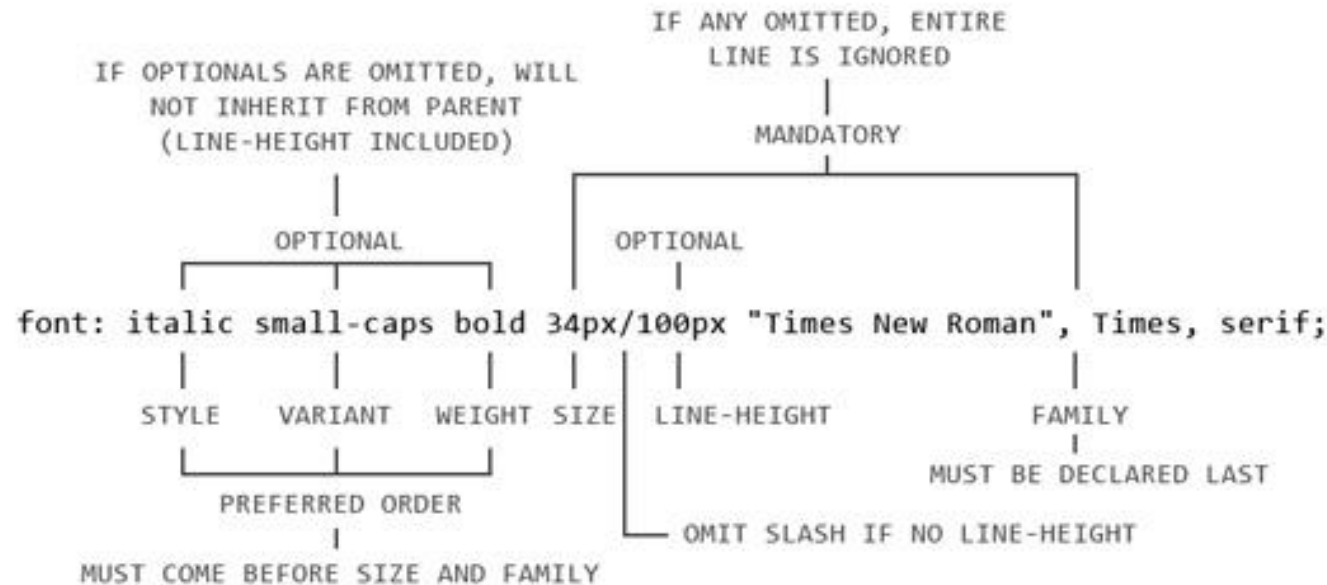
Responsive web design



- Use different styles to fit different displays for the same contents

CSS FONT

CSS FONT SHORTHAND PROPERTY CHEAT SHEET by IMPRESSIVE WEBS



```
<!DOCTYPE html>
<html>
<head>
<style>
p.a {
  font-style: normal;
}
p.b {
  font-style: italic;
}
h1 {
  color: red;
}
</style>
</head>
<body>
<h1>Font styling</h1>
<p class="a">This is a normal paragraph.</p>
<p class="b">This is an italic paragraph</p>
```

Font styling

This is a normal paragraph.

This is an italic paragraph

CSS FONT alignment

CSS

```
/* Keyword values */
text-align: start;
text-align: end;
text-align: left;
text-align: right;
text-align: center;
text-align: justify;
text-align: justify-all;
text-align: match-parent;
```

Integer elementum massa at nulla placerat varius. Suspendisse in libero risus, in interdum massa. Vestibulum ac leo vitae metus faucibus gravida ac in neque. Nullam est eros, suscipit sed dictum quis, accumsan a ligula.
Integer elementum massa at nulla placerat varius. Suspendisse in libero risus, in interdum massa. Vestibulum ac leo vitae metus faucibus gravida ac in neque. Nullam est eros, suscipit sed dictum quis, accumsan a ligula.
Integer elementum massa at nulla placerat varius. Suspendisse in libero risus, in interdum massa. Vestibulum ac leo vitae metus faucibus gravida ac in neque. Nullam est eros, suscipit sed dictum quis, accumsan a ligula.
Integer elementum massa at nulla placerat varius. Suspendisse in libero risus, in interdum massa. Vestibulum ac leo vitae metus faucibus gravida ac in neque. Nullam est eros, suscipit sed dictum quis, accumsan a ligula.
Integer elementum massa at nulla placerat varius. Suspendisse in libero risus, in interdum massa. Vestibulum ac leo vitae metus faucibus gravida ac in neque. Nullam est eros, suscipit sed dictum quis, accumsan a ligula.
Integer elementum massa at nulla placerat varius. Suspendisse in libero risus, in interdum massa. Vestibulum ac leo vitae metus faucibus gravida ac in neque. Nullam est eros, suscipit sed dictum quis, accumsan a ligula.

CSS styling of Link (URL)

```
<head>
  <style type="text/css">
    a:link {
      color:red;
    }
    a:visited {
      color:green;
    }
    a:hover {
      color:blue;
    }
    a:active {
      color:orange;
    }
  </style>
</head>

<a href="http://www.google.com">
  Link to Google
</a>
```

Link to Google

Link to Google

Link to Google

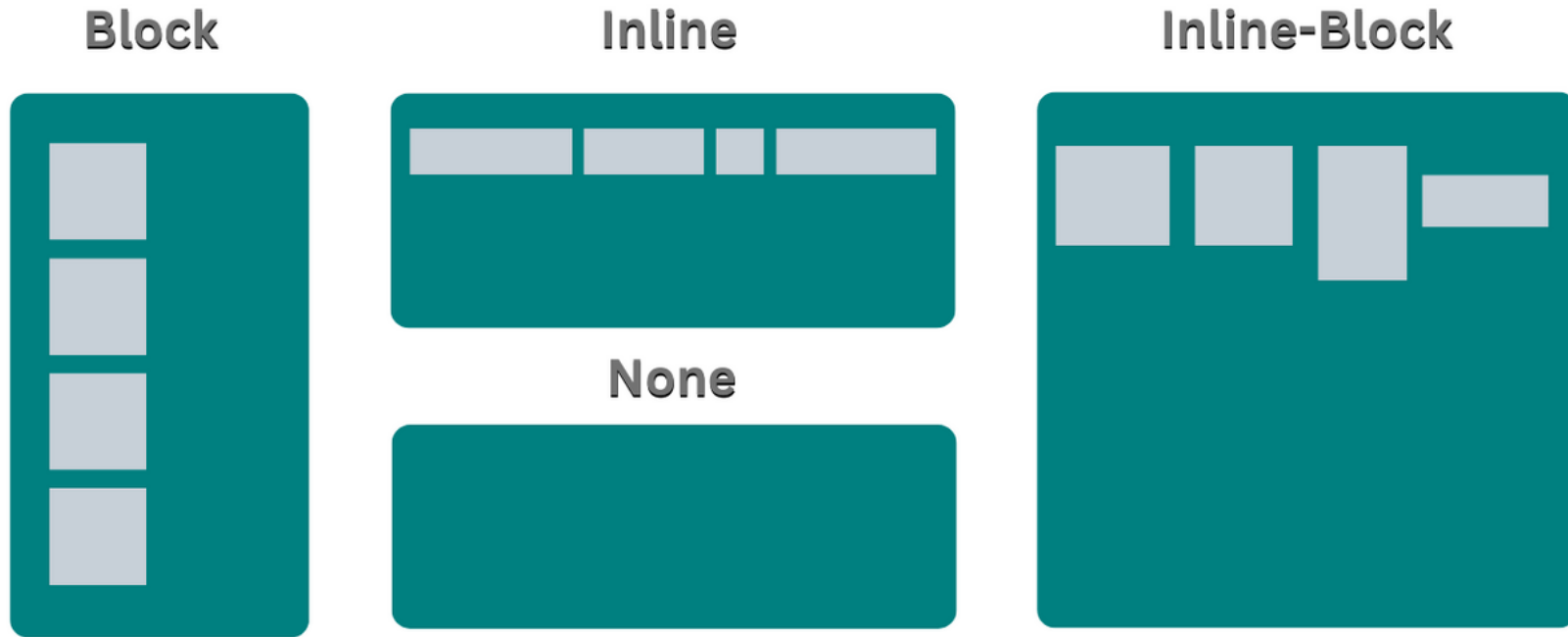
Link to Google

After the link was clicked and we returned to this page.

As we are actually clicking on the link.

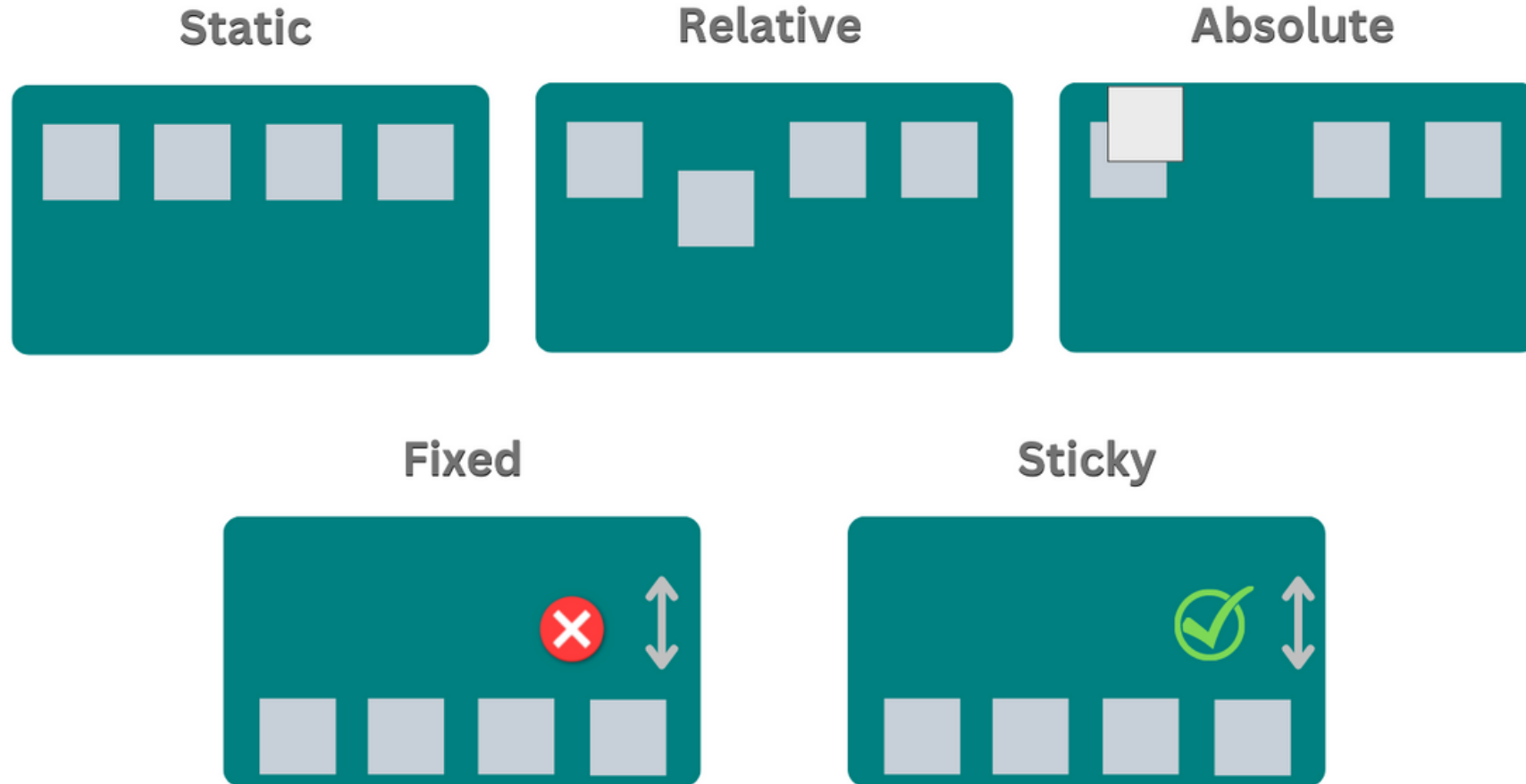
Note: Always list these properties and states in the order shown. Mixing their order on your style sheet can result in unpredictable results.

CSS display



- **This is about the layout between different elements**

CSS position



- **This is about the layout between different elements**

<https://codemafias.com/CSS-Display-&-Position>

CSS...

Boxes

margin *
margin-top
margin-right
margin-bottom
margin-left

padding *
padding-top
padding-right
padding-bottom
padding-left

border *
border-top *
border-bottom *
border-right *
border-left *

border-color *
border-top-color
border-right-color
border-bottom-color
border-left-color

border-style *
border-top-style
border-right-style
border-bottom-style
border-left-style

border-width *
border-top-width
border-right-width
border-bottom-width
border-left-width

Positioning

display
position
top
right
bottom
left
float
clear



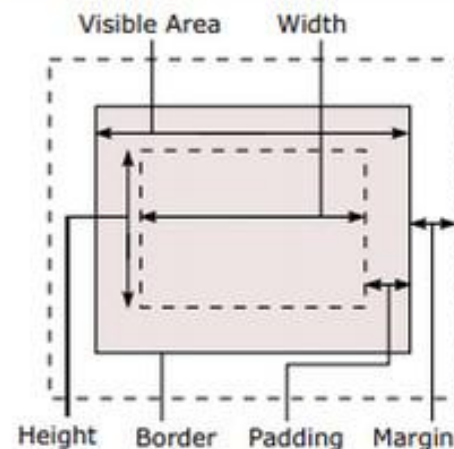
SYNTAX

```
/* Comment */  
@media type {  
  selector {  
    property: values;  
  }  
}  
  
(Media type optional)
```

SELECTORS

	Styles apply to:
*	All elements
div	<div>
div *	Elements within <div>
div span	 within <div>
div, span	<div> and
div > span	 with <div> as parent
div + span	 preceded by ...
.class	Elements of class "class"
div.class	<div> of class "class"
#itemid	Element with id "itemid"
div#itemid	<div> with id "itemid"
a[class]	<a> with class attribute
a[class='x']	<a> when class is "x"
a[class~='x']	<a> when class is a list of space-separated values and one of those is 'x'
a[lang = 'en']	<a> when lang begins

BOX MODEL



UNITS

px	Pixels
em	1em equal to font size of parent (same as 100%)
ex	Height of lower case "x"
%	Percentage
in	Inches
cm	Centimeters
mm	Millimeters
pt	1pt = 1/72in
pc	1pc = 12pt
#789abc	RGB Hex Notation
#acf	Equates to "#aaccff"
rgb(0,25,50)	Value (0 to 255) of each

Paging

size
marks
page-break-before
page-break-after
page-break-inside
page
orphans
widows

Color / Background

color
* background
background-color
background-image
background-repeat
background-attachment
background-position

Fonts

* font
font-family
font-style
font-variant
font-weight
font-stretch
font-size
font-size-adjust

Text

text-indent
text-align
text-decoration
text-shadow
letter-spacing
word-spacing
text-transform
white-space

Tables

caption-side
table-layout
border-collapse
border-spacing
empty-cells
speak-header

Summary

- Basic ideas of HTML - **DOM**
- Basic ideas of CSS syntax

